care centres. Therapeutic exercises are carried out by physiotherapists. The common problem at the end of fracture treatment is the wasting of muscles and stiffness of joints. These problems can be restored by the physiotherapy with gradual exercises. It has proven to be effective in the post surgery treatment and management of arthritis, spondylosis, musculo skeletal disorders, stroke and spinal cord injury.

Summary

Movement is one of the significant features of living organisms. The different types of movements are amoeboid movement, ciliary movement, flagellar movement and muscular movement. Three types of muscles are present in human beings. They are the skeletal muscle, visceral muscle and cardiac muscle. The skeletal muscles are attached to the bones by tendons.

The most striking microscopic feature of skeletal muscle is a series of light and dark bands. The muscles exhibit the properties such as excitability, contractibility, conductibility and elasticity. There are two types of muscle contraction. They are isotonic and isometric contractions.

The skeletal system consists of a frame work of bones and cartilages. The skeletal system is grouped into two principal divisions: the axial skeleton and the appendicular skeleton. There are three types of joints present in the body: fibrous, cartilaginous and synovial joints.

The disorders related to muscular system are myasthenia gravis, muscular dystrophy, tetany, muscle fatigue, muscle pull, atrophy and rigor mortis. The disorders of the skeletal system are arthritis and osteoporosis. Regular body exercise keeps the body fit and healthy.

A typical long bone has a diaphysis (shaft), epiphyses (singular-epiphysis) and membranes. Even though the bones are strong, they are also susceptible to fractures or breaks.There are four major stages in repairing a simple fracture.

Physiotherapy is the therapeutic exercise to make the limbs work near normally.

Evaluation

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1. Muscles are derived from
   a. ectoderm
   b. mesoderm
   c. endoderm d. neuro ectoderm
2. Muscles are formed by
   a. myocytes b. leucocytes
   c. osteocytes d. lymphocytes
3. The muscles attached to the bones are called
   a. skeletal muscle
   b. cardiac muscle
   c. involuntary muscle
   d. smooth muscles
4. Skeletal muscles are attached to the bones by
   a. tendon b. ligament
   c. pectin d. fibrin
5. The bundle of muscle fibres is called
   a. Myofibrils b. fascicle
   c. sarcomere d. sarcoplasm
6. The pigment present in the muscle fibre to store oxygen is
   a. myoglobin b. troponin
   c. myosin d. actin
7. The functional unit of a muscle fibre is
   a. sarcomere b. sarcoplasm
   c. myosin d. actin

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